Marketing Analytics

Visualization and Data Quality

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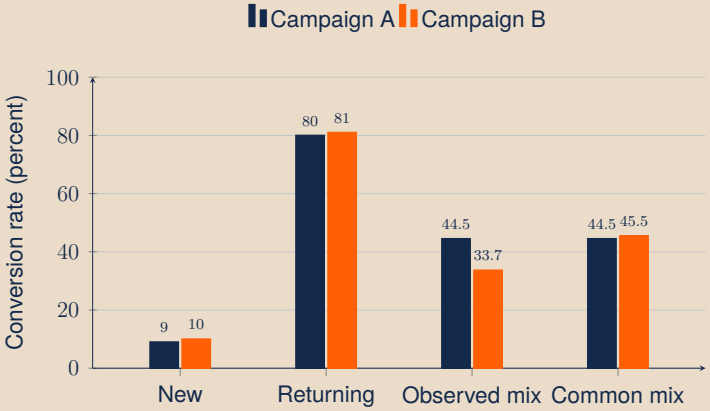
Session 03

A Campaign Comparison

- Campaign B converts more new and returning customers within each group.
- Its aggregate conversion rate is lower.
- The customer mix determines which comparison the aggregate makes.

Customer Mix Reverses the Ranking



Synthetic campaigns. Both within-group rates rise by one percentage point.

A Standardized Conversion Rate

Use common customer-group weights w_g for each campaign c.

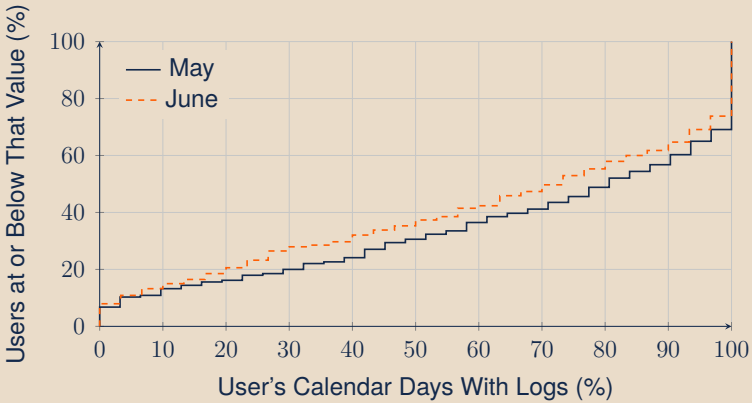
$$\bar{p}_c^{\text{std}} = \sum_g w_g p_{cg}, \qquad \sum_g w_g = 1$$

Changing the weights changes the population represented by the rate.

Chart Choice Follows the Question

- Aligned positions compare magnitudes.
- Lines compare changes over ordered time.
- Distributions reveal spread and concentration.
- Paired displays preserve within-customer changes.

A Mean Decline Can Hide a Stable Median

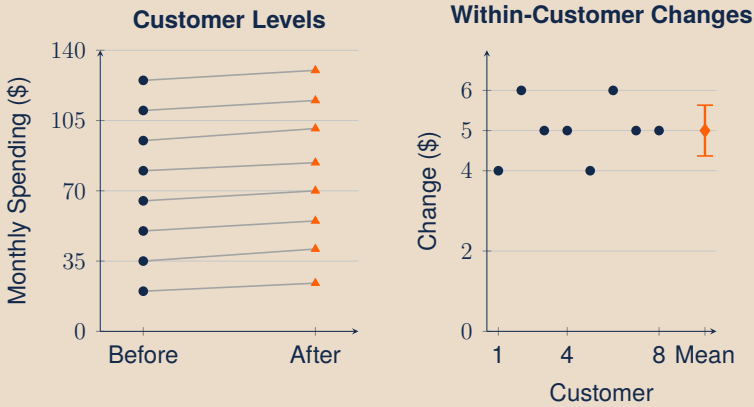


mHealth comparison uses paired customers and a common observation window.

Uncertainty and Dependence

- Repeated days for one customer share common influences.
- Customer-level resampling preserves each customer history.
- A confidence interval and a percentile range describe different objects.

Spending Levels and Customer Changes



Eight synthetic customers; mean change is 5 dollars. Conditional 95% t interval: [4.368, 5.632].

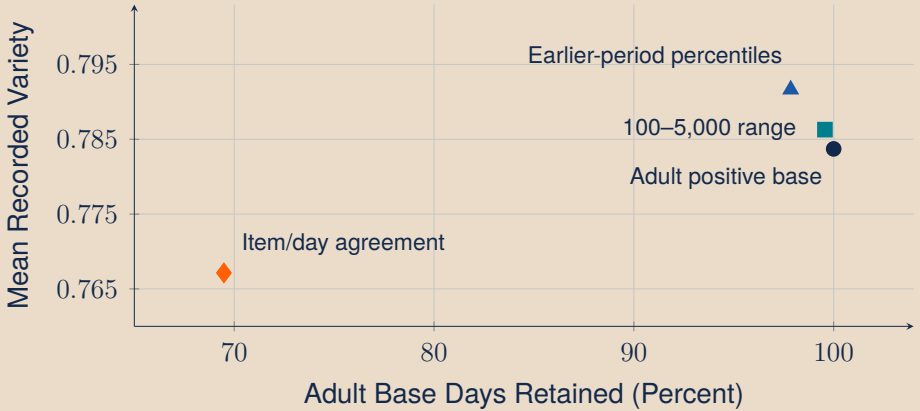
The Sampling Unit Determines Uncertainty

- Preserve customer pairs and compare equal observation windows.
- Within-customer covariance can make a mean change precise.
- A before/after change needs a counterfactual design for a causal claim.

Nutrition Knowledge and Measurement

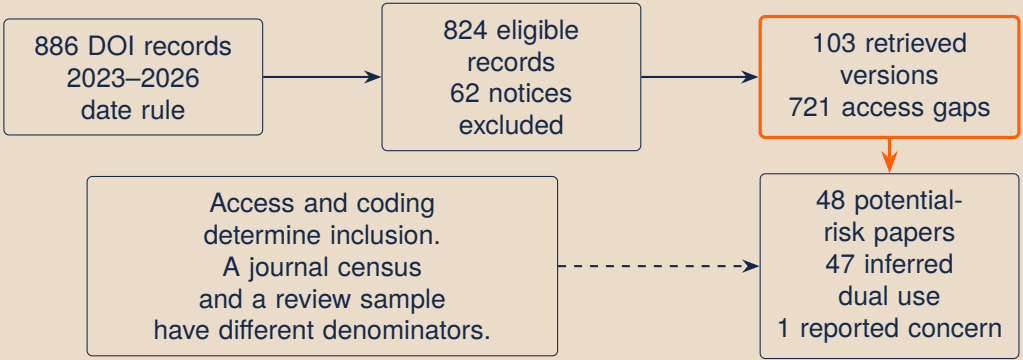
- A recorded food log can omit meals and portions.
- Energy plausibility depends on the measurement instrument and physiology.
- Age, body size, activity, and recording duration affect screening.

Sample Restrictions Change the Comparison



mHealth screening alternatives are explicit analysis choices.

Access Changes the Evidence Sample



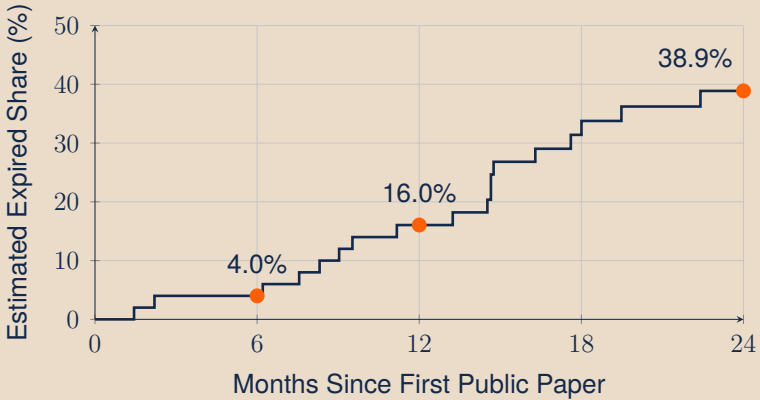
November 5, 2026 snapshot. Counts refer to DOI records and retrieved versions.

Selected Findings at a Fixed Date

Published Category	Claims
Expired	22
Weakened	17
Persists	8
Strengthened	3

Fifty selected claims, twenty models, and a September 2026 endpoint.

Follow-up Changes the Denominator



All fifty claims contribute follow-up; 28 remain right-censored.

Energy-Plausibility Screens

- An intake-to-requirement comparison needs defined inputs.
- Unobserved physiological inputs remain missing.
- A hypothetical Goldberg calculation teaches the method separately from the logs.

Interactive Comparisons

- Keep comparable panels on common scales.
- Display the eligible population and current selection.
- Preserve customers with no recorded days when the estimand includes them.

A Composition Check

Synthetic within-group rates are 9%, 80%, 10%, and 81%.

```
rates_a = [0.09, 0.80]
rates_b = [0.10, 0.81]
common_a = sum(rates_a) / 2
common_b = sum(rates_b) / 2
observed_b = (2 * rates_b[0] + rates_b[1]) / 3
assert round(common_b - common_a, 2) == 0.01
```

B improves by one point under common weights despite its lower observed aggregate.

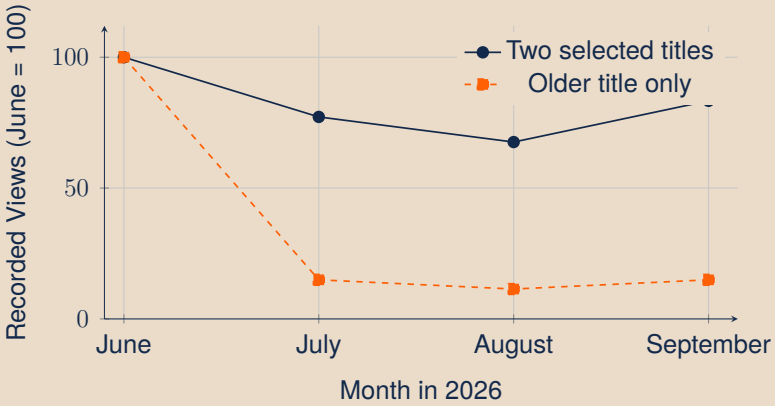
Coding Exercise: A Chart Assessment

- Generate a rate chart from a verified aggregate table.
- Alter the axis range, cohort filter, and denominator one at a time.
- Explain which changes alter the substantive comparison.

A Public-Signal Measurement Problem

- A product can acquire a new page title while demand stays unchanged.
- A single-title series then misses part of its recorded attention.
- Coverage, redirects, and missing dates require an assessment.

Title Changes and Apparent Declines



The supplied public pageview capture uses observed title histories.

Evidence in a Management Presentation

- State the decision and supported result.
- Show the comparison, units, and denominator.
- Keep the qualification that could change the action beside the claim.

Honest Comparisons

- Keep denominators and scales visible.
- Preserve missing outcomes and relevant subgroup variation.
- Avoid presenting a filtered descriptive pattern as a general causal effect.

Visualization Discussion

- Which customers receive weight in the chart?
- Could sample restrictions reverse the recommendation?
- What does the uncertainty interval include?